

Grade 11 – Term 1

C3 Practice Activity 3 with Solutions

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Evaluated criterion

- C3: Constructs uniform and triangular probability density functions for periodic variables and calculates context-specific tail and band probabilities geometrically.

1 Introduction

Problems:

- 1.1. Periodic variables and the cycle cut
 - 1.2. Uniform and triangular pdf shapes
 - 1.3. Nine-step geometric method
-

Criterion C2 constructed continuous pdfs and found probabilities from areas. Its uniform pdfs used rectangles, while its triangular pdfs used triangles and trapezoids. Criterion C3 applies exactly those geometric methods to variables that repeat. No new area method is needed; the new idea is to respect the cycle cut when an event passes from the end of one displayed cycle to the beginning of the next.

The word *circular* describes the **variable**: its values repeat after one period. The words *uniform* and *triangular* describe the **shape of its pdf**. A circular-uniform model therefore has a periodic variable and a rectangular pdf on one displayed cycle. A circular-triangular model has a periodic variable and straight sides that form one physical triangular peak around the cycle.

1.1 Periodic variables and the cycle cut

Compass direction repeats after one revolution, time of day repeats after one day, and rotation repeats when a device returns to its starting orientation. For example,

$$0^\circ \equiv 360^\circ, \quad 0 \text{ h} \equiv 24 \text{ h}, \quad f(x + P) = f(x),$$

where P is the period length. For radians, $P = 2\pi$; for degrees, $P = 360^\circ$; and for time of day in hours, $P = 24$ h.

Displaying one circle as an interval requires a *cut*. For example, $[0^\circ, 360^\circ)$ cuts the compass at north. The interval has a written beginning and end, but the physical compass has no endpoint. Thus 359° and 1° are only 2° apart across the cut. Likewise, 23:59 and 00:01 are close times even though their recorded numbers lie on opposite sides of the midnight cut.

The displayed interval is a recording convention, not a statement that the model stops beyond its endpoints. Consequently, do not automatically add an “otherwise” branch with zero density. Write the pdf on one chosen cycle and state its periodic extension. The equivalent endpoints describe the same physical value, and a wrapped event must be split only for area bookkeeping.



1.2 Uniform and triangular pdf shapes

The only area formulas required in this activity are

$$A_{\text{rectangle}} = bh, \quad A_{\text{triangle}} = \frac{1}{2}bh, \quad A_{\text{trapezoid}} = \frac{1}{2}(h_1 + h_2)b.$$

Here b is horizontal width and each h is a pdf height. A probability is the area of its shaded region, and the area over one complete cycle is 1.

For a uniform pdf on a period of length P , the whole cycle is a rectangle:

$$\text{Uniform: } 1 = Ph, \quad h = \frac{1}{P}.$$

Equal-width arcs or time bands therefore have equal probabilities. This is a reasonable model when a controlled randomizer deliberately gives every part of a cycle the same chance, or when the model is explicitly introduced as a baseline for comparison. Periodicity alone does not imply uniformity.

For a triangular pdf on a period of length P , the complete physical peak has base P and height H :

$$\text{Triangular: } 1 = \frac{1}{2}PH, \quad H = \frac{2}{P}.$$

If the mode lies inside the displayed interval, the graph rises to H and falls to zero at the cut. Straight-side heights can be found from line equations, similar triangles, or proportions, exactly as in C2.

If the physical peak lies at the cut, the unwrapped graph shows two triangular pieces: one descends from the peak at the left endpoint, and the other rises to the same peak at the equivalent right endpoint. If their bases are b_1 and b_2 , then $b_1 + b_2 = P$ and

$$\underbrace{\frac{1}{2}b_1H}_{\text{first displayed piece}} + \underbrace{\frac{1}{2}b_2H}_{\text{second displayed piece}} = \frac{1}{2}PH = 1.$$

These are not two physical modes. They are the two displayed parts of one peak joined across the cut.



1.3 Nine-step geometric method

Use this routine for every problem.

1. Define the variable, units, period, and chosen cut.
2. Identify the rectangular or triangular shape.
3. Use total area 1 to find unknown heights.
4. Find intermediate heights with line equations, similar triangles, or proportions.
5. Write the pdf on one cycle and state its periodic extension.
6. Shade the requested region.
7. Split at the cycle cut or pdf breakpoints.
8. Add geometric areas, or subtract simple outside regions from 1.
9. Check $0 \leq P(\text{event}) \leq 1$ and interpret the result.

A circle has no intrinsic left or right tail. The context must name the direction and the cut. For example, “the clockwise sector from 300° to the end of the displayed cycle” means $[300^\circ, 360^\circ)$. A band within 30° of the cut is written as the wrapped event

$$[330^\circ, 360^\circ) \cup [0^\circ, 30^\circ).$$

The two written pieces are disjoint, so their labelled geometric areas are added.



2 Circular Uniform Distributions

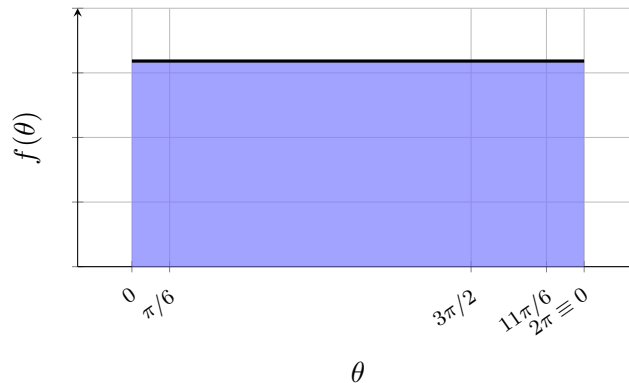
Problems:

- 2.1. Random sensor orientation
 - 2.2. Random daily system-check time
 - 2.3. Randomized rotating-device direction
-

2.1 Random sensor orientation

An optical sensor can point in any orientation. Let Θ be its clockwise orientation from a reference mark, measured in radians and recorded on $[0, 2\pi)$. Orientation is periodic because one full rotation returns the sensor to the same physical direction, so 0 and 2π are equivalent.

During calibration, software generates a direction by controlled randomization that gives equal chances to equal-width arcs. A circular-uniform pdf is therefore reasonable by design, rather than being assumed from natural data. Engineers use the clockwise sector from $3\pi/2$ to the cut to estimate cable strain, and the wrapped band within $\pi/6$ of the reference mark to plan a high-precision calibration test.



Questions/Tasks.

- C3.1** Construct the complete periodic pdf of Θ from the calibration narrative, using total rectangular area 1 to find its height and stating the period.
- C3.2** Calculate geometrically the cable-strain tail probability for the clockwise sector from $3\pi/2$ to the end of the displayed cycle.
- C3.3** Calculate geometrically the calibration-band probability within $\pi/6$ of the reference mark, crossing the cycle cut.

C3.1 - Construct the periodic pdf.

The variable Θ is orientation in radians, its period is 2π , and its one-cycle support representation is $[0, 2\pi)$ with the cut at the reference mark. Controlled equal-arc randomization makes the pdf a rectangle of unknown height h . Its complete-cycle area gives

$$1 = A_{\text{rectangle}} = (2\pi)h, \quad h = \frac{1}{2\pi}.$$

Thus, on the chosen cycle,

$$f_{\Theta}(\theta) = \frac{1}{2\pi}, \quad 0 \leq \theta < 2\pi,$$

and the complete periodic model is stated by

$$f_{\Theta}(\theta + 2\pi) = f_{\Theta}(\theta).$$

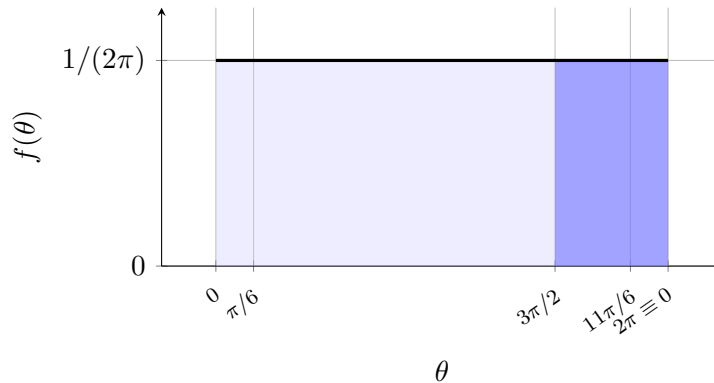
The period is 2π , and the geometric check is $(2\pi)(1/(2\pi)) = 1$.

C3.2 - Calculate the clockwise tail.

Relative to the cut at $0 \equiv 2\pi$, the cable-strain tail means the clockwise sector $[3\pi/2, 2\pi)$. The shaded region below is one rectangle of width $2\pi - 3\pi/2 = \pi/2$ and height $1/(2\pi)$:

$$\begin{aligned} P\left(\frac{3\pi}{2} \leq \Theta < 2\pi\right) &= \underbrace{\left(2\pi - \frac{3\pi}{2}\right)}_{A_{\text{tail rectangle}}} \frac{1}{2\pi} \\ &= \frac{1}{4} = 25\%. \end{aligned}$$

The probability is valid because it lies between 0 and 1. One quarter of randomized calibrations are expected in the cable-strain sector.



C3.3 - Calculate the wrapped calibration band.

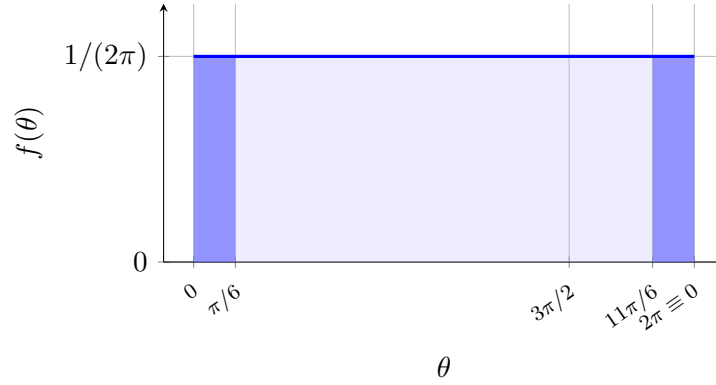
The band crosses the cut, so on $[0, 2\pi)$ it is

$$\left[\frac{11\pi}{6}, 2\pi\right) \cup \left[0, \frac{\pi}{6}\right].$$

The graph shades both written pieces. Each is a rectangle of width $\pi/6$:

$$\begin{aligned} P(\text{calibration band}) &= \underbrace{\frac{\pi}{6} \frac{1}{2\pi}}_{A_1=1/12} + \underbrace{\frac{\pi}{6} \frac{1}{2\pi}}_{A_2=1/12} \\ &= \frac{1}{6} \approx 16.7\%. \end{aligned}$$

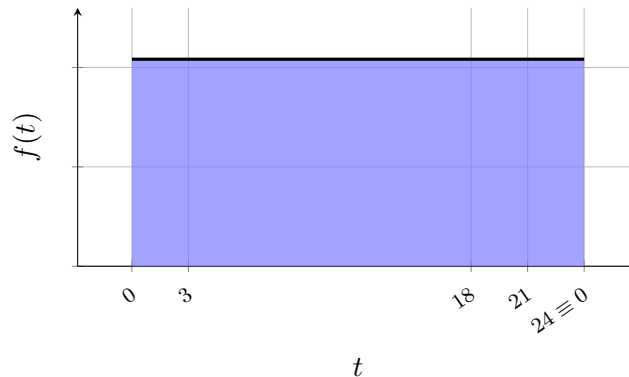
Thus about 16.7% of randomized orientations fall within $\pi/6$ of the reference direction, using both sides of the artificial cut.



2.2 Random daily system-check time

A cloud platform runs one system check each day. Let T be the check's time of day in hours, recorded on $[0, 24)$ with the cut at midnight. Time of day is periodic because the 24-hour clock returns to the same time after one day: 0 h and 24 h are equivalent.

To distribute network load fairly, the scheduler uses controlled randomization that gives equal chances to equal-length time windows. This makes a circular-uniform model reasonable. Operations managers need the late-cycle probability from 18:00 to midnight to plan evening on-call staff, and the overnight probability from 21:00 through 03:00 to decide whether a second overnight response team is justified.



Questions/Tasks.

- C3.1** Construct the complete periodic pdf of T from the scheduler narrative, using total rectangular area 1 to find its height and stating the period.
- C3.2** Calculate geometrically the evening on-call tail probability from 18:00 to the end of the displayed day.
- C3.3** Calculate geometrically the overnight planning-band probability from 21:00 through 03:00, crossing midnight.

C3.1 - Construct the periodic pdf.

The variable T is time of day in hours, its period is 24 h, and its one-cycle support representation is $[0, 24)$ with a midnight cut. Equal-window randomization gives a rectangular pdf of height h . Total area 1 gives

$$1 = A_{\text{rectangle}} = (24)h, \quad h = \frac{1}{24}.$$

On the chosen recording cycle,

$$f_T(t) = \frac{1}{24}, \quad 0 \leq t < 24,$$

and its periodic extension is

$$f_T(t + 24) = f_T(t).$$

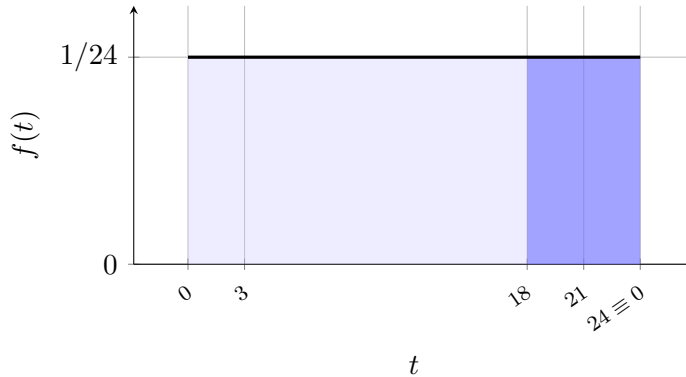
The period is 24 h and the rectangular area check is $24(1/24) = 1$.

C3.2 - Calculate the late-cycle tail.

The evening tail is defined relative to the midnight cut as $[18, 24)$, moving forward in clock time to the end of the displayed day. Its shaded shape is a width-6 rectangle:

$$P(18 \leq T < 24) = \underbrace{(24 - 18)}_{A_{\text{tail rectangle}}} \frac{1}{24} = \frac{1}{4} = 25\%.$$

Therefore one quarter of randomized daily checks require evening on-call coverage.



C3.3 - Calculate the overnight band.

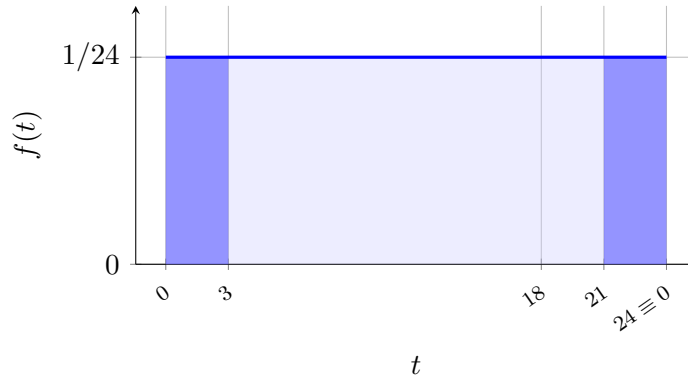
The time band crosses midnight, so the one-cycle representation is

$$[21, 24) \cup [0, 3].$$

The split occurs at the cycle cut. The shaded event consists of two width-3 rectangles:

$$\begin{aligned}
 P(21:00 \text{ through } 03:00) &= \underbrace{(24 - 21)\frac{1}{24}}_{A_1=1/8} + \underbrace{(3 - 0)\frac{1}{24}}_{A_2=1/8} \\
 &= \frac{1}{4} = 25\%.
 \end{aligned}$$

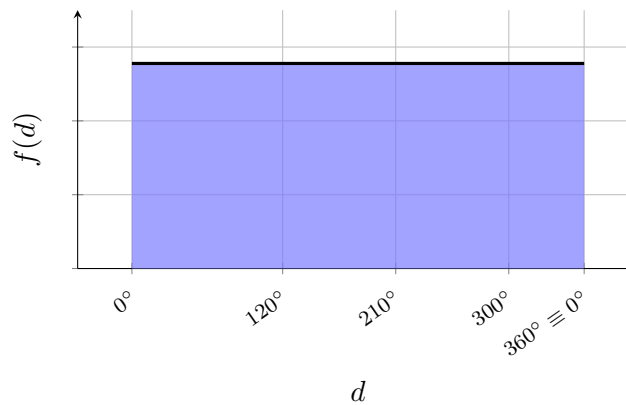
Thus the overnight team would cover 25% of the platform's randomized system checks.



2.3 Randomized rotating-device direction

A rotating quality-control camera records its direction D in degrees clockwise from its home mark on $[0^\circ, 360^\circ)$. Direction is periodic: after a 360° rotation the camera has returned to its home direction, so the displayed endpoints are equivalent.

For each inspection the controller deliberately randomizes the direction with no preferred sector, making a circular-uniform model appropriate by design. The clockwise sector from 300° to the cut places a cable near its safe bend limit, while the 120° – 210° target sector views a weld. The two probabilities support cable-maintenance and target-visibility decisions.



Questions/Tasks.

C3.1 Construct the complete periodic pdf of D from the controller narrative, using total rectangular area 1 to find its height and stating the period.

C3.2 Calculate geometrically the cable-limit tail probability for the clockwise sector from 300° to the end of the displayed cycle.

C3.3 Calculate geometrically the weld-viewing band probability for the target sector from 120° through 210° .

C3.1 - Construct the periodic pdf.

The variable D is direction in degrees, its period is 360° , and its one-cycle support representation is $[0^\circ, 360^\circ)$ with a cut at the home mark. Controlled equal-sector randomization gives a rectangular pdf. If its height is h , then

$$1 = A_{\text{rectangle}} = (360)h, \quad h = \frac{1}{360}.$$

On one displayed cycle,

$$f_D(d) = \frac{1}{360}, \quad 0 \leq d < 360,$$

with density measured per degree. The periodic extension is

$$f_D(d + 360^\circ) = f_D(d).$$

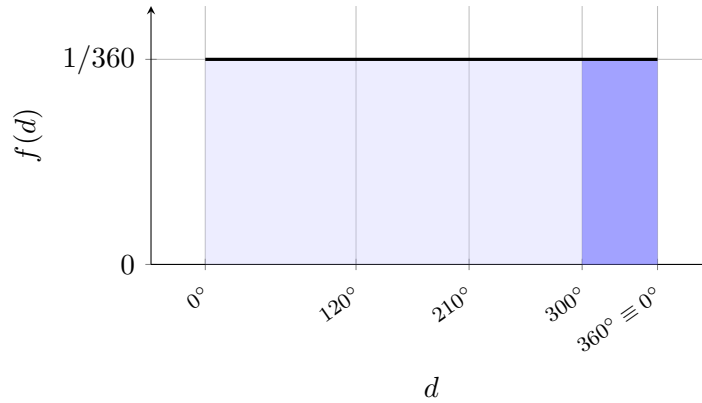
The period is 360° , and $360(1/360) = 1$ verifies the total area.

C3.2 - Calculate the directional tail.

The cable-limit tail is not an intrinsic right tail of a circle. In this context it is specifically the clockwise sector $[300^\circ, 360^\circ)$ from 300° to the home-mark cut. The shaded rectangle has width 60° :

$$P(300^\circ \leq D < 360^\circ) = \underbrace{(360 - 300)}_{A_{\text{tail rectangle}}} \frac{1}{360} = \frac{1}{6} \approx 16.7\%.$$

About 16.7% of inspections place the cable in the maintenance-watch sector.

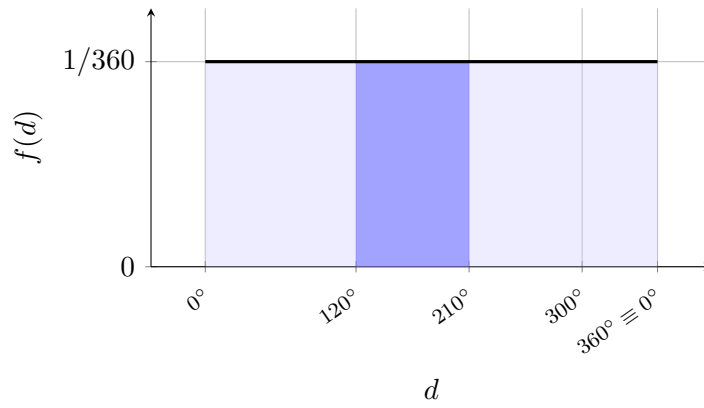


C3.3 - Calculate the target sector.

The target band $[120^\circ, 210^\circ]$ does not cross the cut or a pdf breakpoint. Its shaded region is one width- 90° rectangle:

$$P(120^\circ \leq D \leq 210^\circ) = \underbrace{(210 - 120)}_{A_{\text{target rectangle}}} \frac{1}{360} = \frac{1}{4} = 25\%.$$

The randomized camera views the weld sector in one quarter of inspections, which gives the quality-control team a baseline for expected image coverage.



3 Circular Triangular Distributions

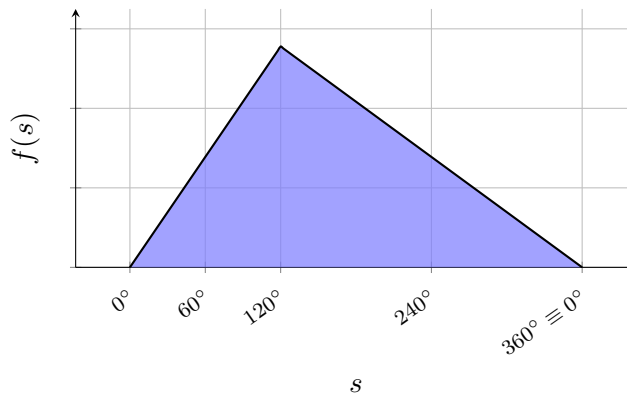
Problems:

- 3.1. Scanner direction with an interior mode
 - 3.2. Daily peak-activity time
 - 3.3. Direction with its peak at the cut
-

3.1 Scanner direction with an interior mode

An automated warehouse scanner records its clockwise direction S in degrees on $[0^\circ, 360^\circ)$, with the cut at north. Direction is periodic because 0° and 360° are the same physical direction.

The programmed scan profile makes north least likely, increases the direction density at a constant rate to its preferred loading-bay direction of 120° , and then decreases it at a constant rate back to zero at the cut. A circular-triangular pdf is reasonable because this controlled profile has one preferred direction and straight changes on both sides. The sector from 240° to the cut measures exposure to a reflective wall, while the 60° – 240° band is the bay-coverage range used to position stock.



Questions/Tasks.

- C3.1** Construct the complete periodic pdf of S from the scan profile, using total triangular area 1 to find all heights and stating the period.

C3.2 Calculate geometrically the reflective-wall tail probability for the clockwise sector from 240° to the end of the displayed cycle.

C3.3 Calculate geometrically the bay-coverage band probability from 60° through 240° .

C3.1 - Construct the periodic pdf.

The variable S is direction in degrees, the period is 360° , and its one-cycle support representation is $[0^\circ, 360^\circ)$ with its cut at north. The graph is a triangle with mode 120° , zero height at the cut, and unknown peak H . Total area 1 gives

$$1 = A_{\text{triangle}} = \frac{1}{2}(360)H, \quad H = \frac{1}{180}.$$

The rising line through $(0, 0)$ and $(120, 1/180)$ has slope

$$\frac{1/180 - 0}{120 - 0} = \frac{1}{21600},$$

so its equation is $s/21600$. The falling line through $(120, 1/180)$ and $(360, 0)$ has slope $-1/43200$; using $(360, 0)$ gives $(360 - s)/43200$. Therefore, on one cycle,

$$f_S(s) = \begin{cases} \frac{s}{21600}, & 0 \leq s \leq 120, \\ \frac{360 - s}{43200}, & 120 < s < 360. \end{cases}$$

Its periodic extension is

$$f_S(s + 360^\circ) = f_S(s).$$

The areas verify the model geometrically:

$$\underbrace{\frac{1}{2}(120)\frac{1}{180}}_{1/3} + \underbrace{\frac{1}{2}(240)\frac{1}{180}}_{2/3} = 1.$$

C3.2 - Calculate the reflective-wall tail.

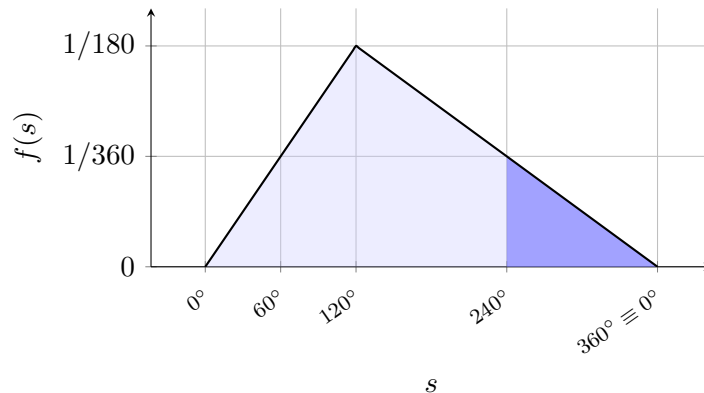
Relative to the north cut, this tail is the clockwise sector $[240^\circ, 360^\circ)$. The line equation gives

$$f_S(240) = \frac{360 - 240}{43200} = \frac{1}{360}.$$

The shaded tail is a triangle ending at zero height at the cut, with base 120° and height $1/360$:

$$P(240^\circ \leq S < 360^\circ) = \underbrace{\frac{1}{2}(360 - 240)\frac{1}{360}}_{A_{\text{tail triangle}}} = \frac{1}{6} \approx 16.7\%.$$

Thus the scanner points toward the reflective-wall sector on about 16.7% of programmed scans.



C3.3 - Calculate the bay-coverage band.

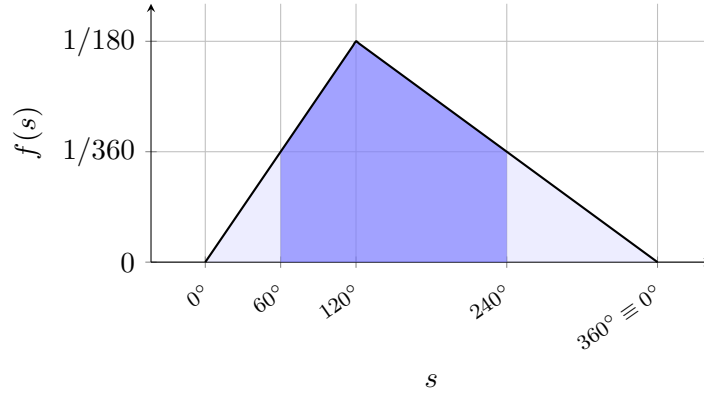
The band $[60^\circ, 240^\circ]$ crosses the pdf breakpoint at the mode, so it is split at 120° . The boundary heights are

$$f_S(60) = \frac{1}{360}, \quad f_S(120) = \frac{1}{180}, \quad f_S(240) = \frac{1}{360}.$$

The two shaded pieces are trapezoids:

$$\begin{aligned} P(60^\circ \leq S \leq 240^\circ) &= \underbrace{\frac{1}{2} \left(\frac{1}{360} + \frac{1}{180} \right) (60)}_{A_1=1/4} \\ &\quad + \underbrace{\frac{1}{2} \left(\frac{1}{180} + \frac{1}{360} \right) (120)}_{A_2=1/2} \\ &= \frac{3}{4} = 75\%. \end{aligned}$$

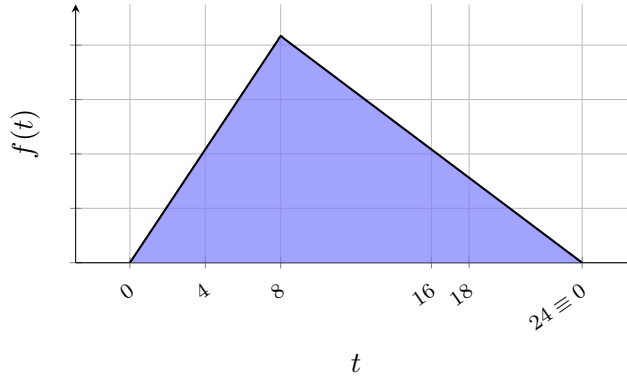
The stock-positioning band covers three quarters of the programmed scan directions.



3.2 Daily peak-activity time

A service desk records T , the time of day at which its daily online activity reaches its peak, in hours on $[0, 24)$. Time of day is periodic because the clock repeats every 24 hours and midnight is both 0 h and 24 h. The desk chooses its cut at midnight, the least likely peak-activity time.

Historical planning summaries show that the density rises at a constant rate from zero at midnight to a mode at 08:00, then falls at a constant rate to zero at the next midnight. A circular-triangular pdf is a simple reasonable model for that single daily preference and those straight changes. A peak after 18:00 requires late support, while the 04:00–16:00 band determines how often the regular planning window contains the day's peak.



Questions/Tasks.

- C3.1** Construct the complete periodic pdf of T from the activity narrative, using total triangular area 1 to find all heights and stating the period.
- C3.2** Calculate geometrically the late-support tail probability from 18:00 to the end of the displayed day.
- C3.3** Calculate geometrically the regular-planning band probability from 04:00 through 16:00.

C3.1 - Construct the periodic pdf.

The variable T is daily peak-activity time in hours. Its period is 24 h, and its one-cycle support representation is $[0, 24)$ with the cut at the least likely time, midnight. The pdf is a triangle with mode 8 and unknown peak H . Its complete-cycle area gives

$$1 = A_{\text{triangle}} = \frac{1}{2}(24)H, \quad H = \frac{1}{12}.$$

The line from $(0, 0)$ to $(8, 1/12)$ has slope $1/96$, so the rising equation is $t/96$. The line from $(8, 1/12)$ to $(24, 0)$ has slope $-1/192$; through $(24, 0)$ its equation is $(24 - t)/192$. On one cycle,

$$f_T(t) = \begin{cases} \frac{t}{96}, & 0 \leq t \leq 8, \\ \frac{24 - t}{192}, & 8 < t < 24. \end{cases}$$

The periodic extension is

$$f_T(t + 24) = f_T(t).$$

The two triangular pieces have total area

$$\underbrace{\frac{1}{2}(8)\frac{1}{12}}_{1/3} + \underbrace{\frac{1}{2}(16)\frac{1}{12}}_{2/3} = 1.$$

C3.2 - Calculate the late-support tail.

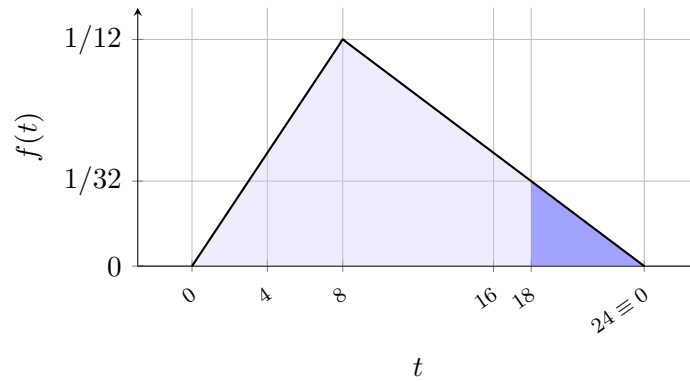
The late-cycle tail is defined as $[18, 24)$, moving forward in clock time from 18:00 to the midnight cut. Its starting height is

$$f_T(18) = \frac{24 - 18}{192} = \frac{1}{32}.$$

The shaded tail is a triangle of base 6 h and height $1/32$:

$$P(18 \leq T < 24) = \underbrace{\frac{1}{2}(24 - 18)\frac{1}{32}}_{A_{\text{tail triangle}}} = \frac{3}{32} = 9.375\%.$$

Therefore about 9.4% of days have a modeled peak late enough to require late-support coverage.



C3.3 - Calculate the regular-planning band.

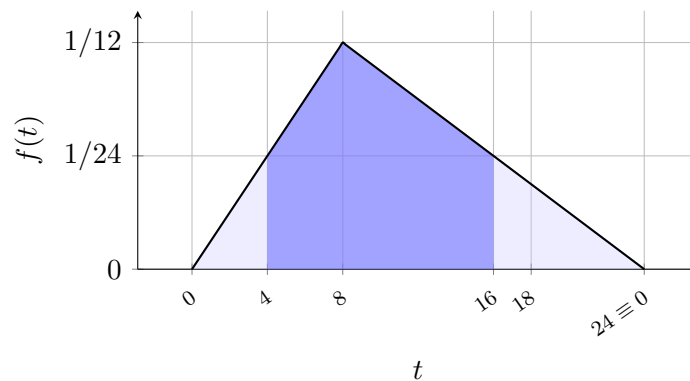
The band $[4, 16]$ crosses the mode at 8, which is a pdf breakpoint. Its three required heights are

$$f_T(4) = \frac{1}{24}, \quad f_T(8) = \frac{1}{12}, \quad f_T(16) = \frac{1}{24}.$$

Splitting the shaded band at 8 produces two trapezoids:

$$\begin{aligned} P(4 \leq T \leq 16) &= \underbrace{\frac{1}{2} \left(\frac{1}{24} + \frac{1}{12} \right) (8 - 4)}_{A_1=1/4} \\ &\quad + \underbrace{\frac{1}{2} \left(\frac{1}{12} + \frac{1}{24} \right) (16 - 8)}_{A_2=1/2} \\ &= \frac{3}{4} = 75\%. \end{aligned}$$

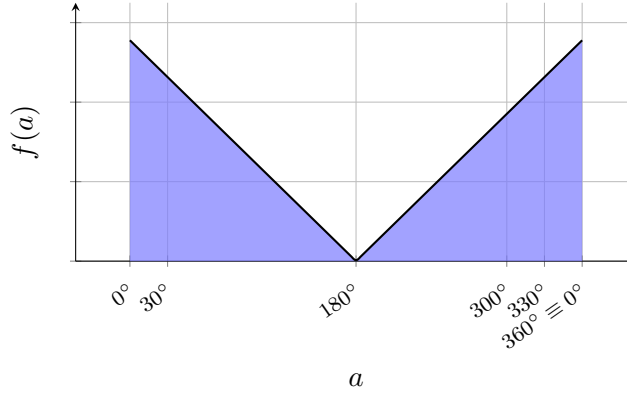
Thus the regular planning window contains the daily activity peak on 75% of modeled days.



3.3 Direction with its peak at the cut

A rotating radio antenna records its direction A in degrees clockwise from north on $[0^\circ, 360^\circ)$. Direction is periodic, so north is both 0° and the equivalent endpoint 360° ; the physical phenomenon does not stop at the recording cut.

The antenna's controller most strongly favors north, reduces the direction density at a constant rate to zero at south (180°), and then increases it at the same constant rate back toward north. A circular-triangular pdf is reasonable for this symmetric programmed preference. The upper-end sector from 300° to the cut tests one cable route, while the band within 30° of north measures whether a receiver can use a narrow target cone.



Questions/Tasks.

- C3.1** Construct the complete periodic pdf of A , showing how two displayed triangular pieces form one physical peak at the cut, using their combined area 1 to find all heights, and stating the period.
- C3.2** Calculate geometrically the cable-route tail probability for the clockwise upper-end sector from 300° to the end of the displayed cycle.
- C3.3** Calculate geometrically the receiver's wrapped target-band probability on $[330^\circ, 360^\circ) \cup [0^\circ, 30^\circ]$.

C3.1 - Construct the periodic pdf.

The variable A is antenna direction in degrees, its period is 360° , and its one-cycle support representation is $[0^\circ, 360^\circ)$ with the cut at the physical mode, north. The unwrapped graph shows a descending triangle on $[0^\circ, 180^\circ]$ and an ascending triangle on $[180^\circ, 360^\circ)$. Their endpoint heights both represent the same northward peak H .

The combined area is 1:

$$1 = \underbrace{\frac{1}{2}(180)H}_{\text{left displayed triangle}} + \underbrace{\frac{1}{2}(180)H}_{\text{right displayed triangle}} = 180H, \quad H = \frac{1}{180}.$$

The first side has slope $-H/180 = -1/32400$ and passes through $(180, 0)$, giving $(180 - a)/32400$. The second has slope $1/32400$ and also passes through $(180, 0)$, giving $(a - 180)/32400$. On one cycle,

$$f_A(a) = \begin{cases} \frac{180 - a}{32400}, & 0 \leq a \leq 180, \\ \frac{a - 180}{32400}, & 180 < a < 360. \end{cases}$$

The complete model repeats according to

$$f_A(a + 360^\circ) = f_A(a).$$

Each displayed triangle has area $1/2$, so their areas sum to 1. The peak at 0° and the height approached at 360° are one physical peak, not two modes.

C3.2 - Calculate the upper-end directional tail.

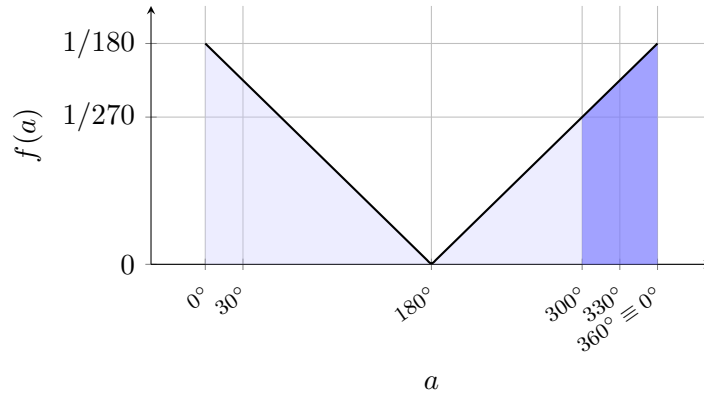
Relative to the north cut, the cable-route tail is the clockwise sector $[300^\circ, 360^\circ)$. Its parallel heights are

$$f_A(300) = \frac{120}{32400} = \frac{1}{270}, \quad f_A(360^-) = \frac{1}{180}.$$

The upper endpoint is excluded from the written cycle, but its approached height completes the geometric edge at the equivalent north direction. The shaded tail is a trapezoid of width 60° :

$$\begin{aligned} P(300^\circ \leq A < 360^\circ) &= \underbrace{\frac{1}{2} \left(\frac{1}{270} + \frac{1}{180} \right)}_{A_{\text{tail trapezoid}}} (60) \\ &= \frac{5}{18} \approx 27.8\%. \end{aligned}$$

Thus about 27.8% of programmed directions use the cable route associated with this upper-end sector.



C3.3 - Calculate the wrapped target band.

The physical target cone is continuous around north, but the chosen recording cycle splits it at the cut:

$$[330^\circ, 360^\circ) \cup [0^\circ, 30^\circ].$$

The outer boundary heights are equal:

$$f_A(330) = \frac{150}{32400} = \frac{1}{216}, \quad f_A(30) = \frac{150}{32400} = \frac{1}{216}.$$

Each shaded piece is a trapezoid of width 30° , with heights $1/216$ and $1/180$ at its parallel sides:

$$\begin{aligned} P(\text{target cone}) &= \underbrace{\frac{1}{2} \left(\frac{1}{216} + \frac{1}{180} \right) (30)}_{A_1=11/72} \\ &\quad + \underbrace{\frac{1}{2} \left(\frac{1}{180} + \frac{1}{216} \right) (30)}_{A_2=11/72} \\ &= \frac{11}{36} \approx 30.6\%. \end{aligned}$$

The result lies between 0 and 1. About 30.6% of programmed antenna directions fall within 30° of north, using both sides of the cut.

